Question 1. Investing in real estate requires a good understanding of how this asset class differs from other financial investments.

a- Discuss the key risks and opportunities of direct real estate investment. (10 points)

Risk and opportunities were discussed in Lecture 1 and touched upon again at several other points in the course. They include vacancy vs. leasing risk, natural disasters, operating costs, liquidity; and rental values, asset appreciation and operational synergies.

b- Clarify why direct real estate investments are intrinsically illiquid compared to other financial assets and discuss the implications of illiquidity in terms of risk and opportunities. (10 points)

This point was discussed in Lecture 1 and then time and again in many other lectures. Good students should also make a connection to large size and use of leverage which further amplifies risks.

c- Leverage can amplify opportunities of real estate investment. Why and under which conditions? (5 points)

These points were discussed in lectures 3 and 4. Good students are expected to support their intuitions by providing a WACC formalisation.

d- What are the costs of leverage (besides interest charges)? (5 points)

This was discussed in Lecture 3 (and 4). The cost of leverage is risk. It would be good if students either formalised this point with a WACC formula or gave an intuition of why leverage creates risk (equity, not debt takes a knock if things go wrong – banks don't really take haircuts).

e- How do you use a discounted cash flow (DCF) approach to model risk and opportunities in real estate investments? Which key elements of your DCF model do you stress to understand risks and opportunities when considering a possible investment? Reflect on what was discussed during the lectures and what you did in the context of your two case studies. (15 points)

This is a fairly open answer that spans many different lectures and includes the case study materials. I would really like students' answer to show that they have engaged with the case study materials so superficial answers will not get top points.

f- Consider now a private equity deal where general partners (GPs) and limited partners (LPs) jointly invest in buying a real estate asset. How are risks and opportunities divided between GPs and LPs? How does a waterfall analysis helps you formalise this 'division' of risk and opportunities? (20 points)

This was the topic of lecture 5 – which included a pre-recorded part by the main lecturer as well as a 1.5hours intervention by Damian Perry. I do not expect any formalisation but a good discussion about the 'pecking' order between LPs and GPs when it comes to remuneration from opportunities and losses (i.e., risk).

g- What is the role of the 'promote' in incentivising efforts by the GP? And what is the difference between a 'deal by deal' and a 'whole fund' approach in remunerating the GP? (15 points)

The promote concept was discussed both in the main lecturer's pre-recorded materials and in the live lecture by Damian Perry. The latter also covered in some details the difference between 'deal by deal' vs. 'whole fund' strategies – and I expect the students to show they have engaged with these concepts as they incentivise GPs differently and are deployed differently in the US vs UK markets.

h- Consider now real estate development. Some of the biggest risks and opportunities come from being able to manage (or not) the construction companies acting on your behalf. What are these key risks and opportunities in this context? And how different contractual agreements (e.g., turnkey vs. forward fund vs. forward commit) allow you manage risks and maximise opportunities? (20 points)

These points were made by the two guest speakers from DFI (Paul and Enrico). They discussed how important is to manage costs and timelines in the process of development. And they highlighted how different contracts with developers (turnkey, forward fund or commit) really affect the likelihood of over-run in terms of costs and time – and which actor is more likely to suffer the consequences (and/or incentivised to avoid problems).

Question 2. You are planning to buy a house that costs £1,000,000 and need a loan for £750,000. Your go to your local bank to discuss a number of mortgage products. They all have a 25-year maturity and monthly compounding. The bank applies a debt coverage ratio of 1.4 and an income multiple of 5.

a- The first loan the bank is offering is an interest only (IO) loan with an interest charge 5.5%. What are the monthly payments for this loan? What will be the value of the last payment (i.e., when the mortgage expires)? (5 points)

Super easy 'warm up' question. The 5.5% interest annually is a 0.4583% monthly. So the monthly payment is 3,437.5 except for the last payment where there is a bullet or balloon repayment that includes the 750,000 original loan.

b- You are also considering a constant payment mortgage (CPM) with an interest rate of 4.9%. What are the monthly payments of this loan? (10 points)

This is also a very straight forward answer. Many such examples were discussed. The data are as follows. Monthly compounding. N=300 – which comes from 25 years at monthly compounding. This should be done by first setting p/y to 12 so the calculator knows compounding. The I/Y is 4.9% and the loan -750,000. This gives 4,340.84. Constant over time.

c- Why is the bank charging a lower interest on the CPM loan than on the IO one? (5 points)

This is because the bank takes less risk – as the IO does not have amortization. This was explicitly discussed several times.

d- The last fixed interest rate loan the bank offers is a Constant Amortization Loan (CAM) with an interest of 4.5%? What is the value of the first payment on this loan? And the value of the payment at the end of the first year? What is the value of the last payment? (15 points)

The key here is that every month the loan is amortizing a constant amount of 750,000/300 (=25*12) = 2500. The first payment is this amount plus interests (4.5%/12=0.375%) on the full amount. This is 2,812.5, so the total first payment is 5,312.5. The 12th payment still has a fixed amount of 2500 but 11 instalments have already been paid so the interest is calculated on 750,000 – 11*2,500 = 722,500 * 0.375% = 2709.3. The total is 5209.375. The last payment has amortized 299 contributions of 2500 so the outstanding loan is 2500 (makes sense: it is one last amortizing payment…). The interest is 9.375 and the overall cost is 2509.375. We discuss CAM in lecture 7 or 8 – and an example was given.

e- Why is the bank charging an even lower rate on the CAM loan than on the CPM and IO mortgages? (5 points)

Same as above: amortization – which is front loaded in CAM compared to CPM. So it's less risky and a lower interest can be charged.

f- Imagine you earn £160,000 per year or £13,333 per month. Given the bank's debt coverage ratio and income multiple rules, are you constrained in terms of your choices? Which mortgage do you prefer and why? (10 points)

Red herring question. The person can take all the loans. If the DCR is calculated on the gross salary of 13,300 even the most expensive one in the initial period (CAM) does not breach the bank's rules. However, CAM is the last preferred – as it does not align with earning potentials. I

am expecting some interesting discussions of CPM vs. IO as more considerations are involved. These issues were discussed in Lectures 7 and 8.

g- You decide to go for the CPM loan and have now been advised there is a £7,500 (1%) origination charge. What is the effective cost of borrowing of this loan if you hold it to maturity? How does your answer change if you repay after 10 years and after 5 years? (20 points)

These are issues we cover in great details in Lectures 7 and 8. The solution is as follows. The payment for the CPM was worked out before. It's 4,340.84. Vis-à-vis this payment, the borrower actually gets 99% of the loan or -742,500. So, this is not the PV and I/Y needs to be worked out with an FV of zero if held to maturity. This gives an IRR of 5.006%. To figure out the effective cost of borrowing at years 10 and 5, students need to use the amortization worksheet to find the outstanding balance. At time 120 (10 years), this is 552,554.93. At time 60 (5 years), this is 663,286.58. Using this as FV and using 10 and 5 for the length of the mortgage, the I/Y can be worked out. The effective cost of borrowing at 10 years is 5.0431%; at 5 years it is 5.1397%.

h- What do you notice in your answer above, and what does this tell you about the reasons why banks charge discount points and other origination charges? (5 points)

The patterns are expected: banks' charges are like a fixed cost that increases the borrowing effective yield the sooner the contract ends. The bank do this to prevent repayment and attract less mobile borrowers. Borrowers that repay do so mostly at times of decreasing rates when they can refinance cheaply. This is a problem that the bank is trying to avoid – and I expect good answers getting full points to make the connection with falling rates.

i- The bank is also offering an Adjustable-Rate Mortgage (ARM). Like the CPM it carries an initial rate of 4.9%. This rate has been fixed for the first three-years. After that, you expect the rate to adjust at 4.5% for two years and then adjust to 5% from start of year 6 onwards. This loan also carries a 1% origination charge, and you expect to repay the loan at the end of Year 10. What is the effective cost of borrowing of this loan? Do you prefer this mortgage, or the CPM loan discussed above (point g – with origination charges)? Why? (20 points)

The first part is like the CPM. So the payment of 4,340.8 still applies to the first three years vis a vis a loan of 750,000 minus 1% commission charge = 742,500. To calculate the new charges at the start of Year 4, the students need the outstanding balance of the original loan at the end of year 3 or 36 periods in. This is 700,533.80. On this outstanding balance, the students will have a 22 years loan (25 years – 3 years) charging 4.5% with a CPM structure. This gives a payment of 4,184.89. For the last part of the loan, the students need the outstanding balance of the previous part after 2 years, that is 24 months in. This is 661,486.79. On this, the students will have to apply a 5% interest rate for the remaining 20 years (from start of Year 6 to maturity). This gives a payment of 4,365.52. At the end of year 10 – meaning five years after the part of the loan charging 5% starts – the outstanding balance is 552,042.89. The effective cost of borrowing can be worked out with the discount cash flow worksheet. The CF0 is -742,500 (not the full amount!). Then the borrower pays 36 periods of 4,340.8. After that, there will be 24 periods of 4,184.89. Then there will be 59 periods of 4,365.52 and in the last period, there will be 4,365.52 plus the outstanding loan at 552,042.89. This gives a monthly IRR of 0.4165% that multiplied by 12 gives 4.980% – if I am not wrong ☺ The cost of borrowing is virtually the same as in g-, just a bit lower. This is because there are periods where the interest rate is lower and others where it's higher. The borrower will

choose one or the other mostly depending on his/her preferences for risk. The little discount the borrower gets ex post is a sort of measure of the risk-premium not charged by the bank for ARM loans compared to CPM. Good answer should try to articulate this point which should be at the heart of the next question.

j- What is the rationale behind ARMs and why do they 'share risks' between banks and borrowers? What are the advantages for borrowers and what are those for banks? (5 points)

These issues were discussed in Lecture 8. ARMs tend to charge lower risk premia as the bank is possibly taking advantage of increasing rates in case of inflationary pressures. It's a fairly open-ended question but I am hoping to see the connection between varying rates, risk premia and benefits for borrowers and banks depending on the real estate cycle.